

Data Analyst Project

Instagram Reach Analysis and Prediction

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Understanding the business problem, objectives, and analytical framework.

Business Understanding

Instagram has become a key platform for media to distribute content and engage audiences. This project analyzes content characteristics, posting patterns, account attributes, and engagement metrics to identify factors influencing post reach and develop machine learning models to predict Instagram reach.

Objectives

- 01
- Identify reach and engagement trends over time and across media profiles.
- 02
- Analyze posting behavior across content types, posting hours, and days of the week to uncover patterns associated with higher reach.
- 03
- Predict Instagram reach through machine learning models and determine the most effective approach.

Framework

Data Understanding	Data Preparation	EDA	Modelling	Evaluation	Deployment
Obtain the Instagram dataset from SocialInsider and explored 3,756 rows x 19 columns.	Process, transform, clean, standardize, and encoded the raw dataset into an analysis-ready format.	Identify content characteristic, engagement trends, and posting behavior.	Split 80%/20% dataset and train Linear Regression, Random Forest, and XGBoost.	Compare R ² , MAE, and RMSE	Recommendations for Instagram reach optimization

Project Overview

Understanding dataset structure, quality, and variables.

Total Rows

3,756

Total Column

19

Total Missing Values

11,314

Total Duplicate Rows

0

Source

SocialInsider

Period

26 May–24 June 2026

Additional Notes

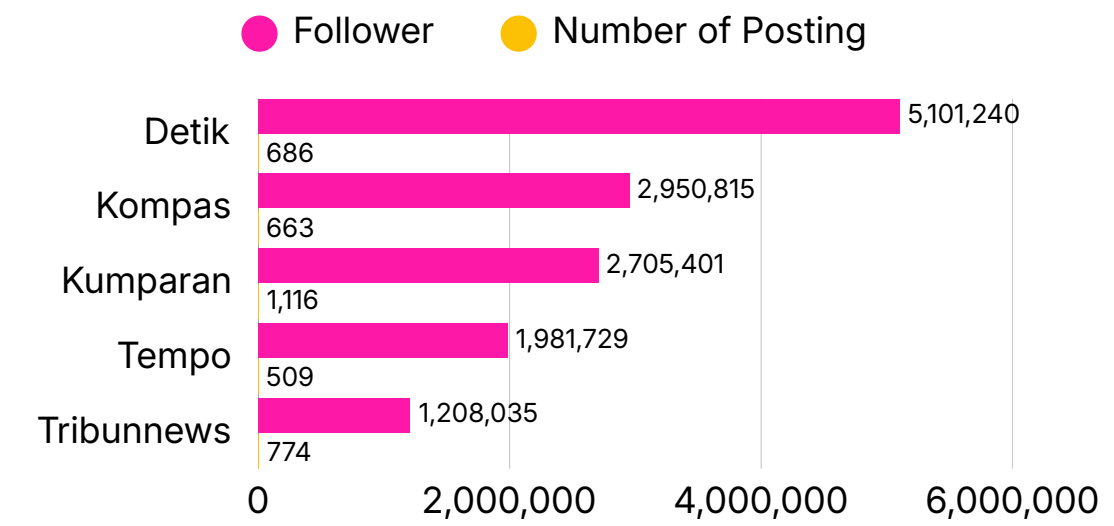
- Engagement was sum of Likes and Comments only.
- Column with all missing values will be removed.
- Unnecessary columns (Brand Content Pillars and Industry Content Pillars) will be removed.
- There will be no topic classification in this project.

Column	Description
Platform	Social media platform where the content was
Profile	Account or profile name that published the content.
Link	Direct URL to the post.
Text	Caption or text content of the post.
Date	Publication date of the post.
Type	Content format (e.g., image, video, reel, carousel).
Engagement	Total interactions received by the post.
Engagement Rate	Engagement relative to followers.
Followers	Number of followers of the publishing account.
Likes	Total likes received by the post.
Comments	Total comments received by the post.
Shares	Total shares received by the post.
Reach	Number of unique users who viewed the post.
Impressions	Total number of times the post was displayed.
Video Views	Total video views received by the post.
Brand Content Pillars	Brand-specific content category or theme.
Industry Content Pillars	Industry-level content category or theme.
Country	Country associated with the account or content.
Unnamed: 18	

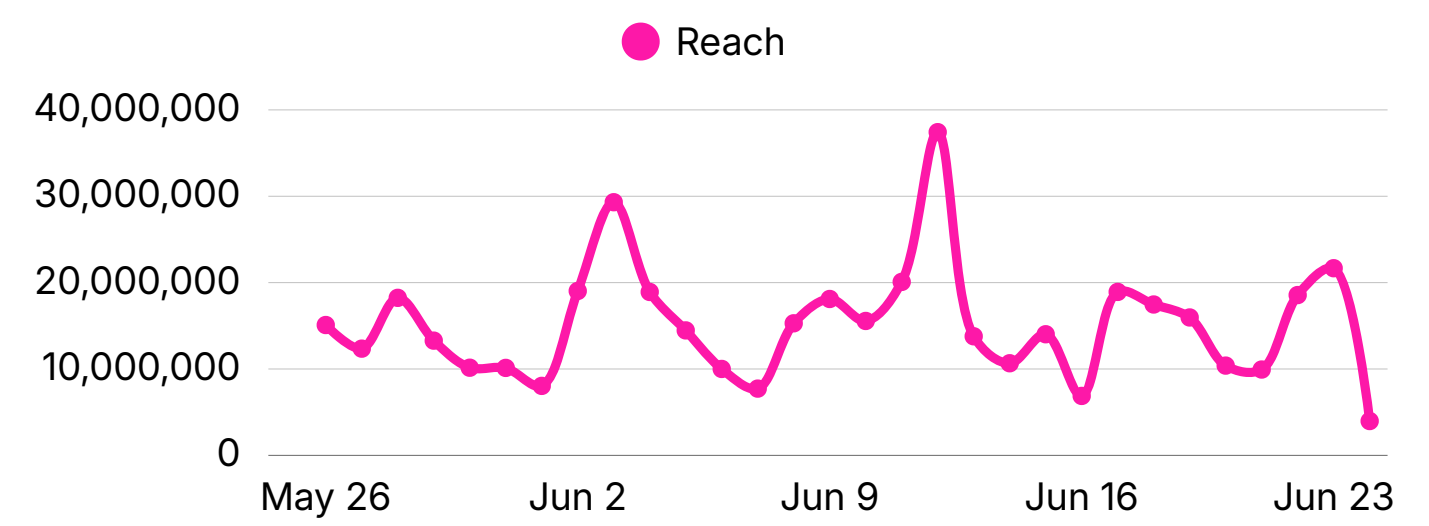
All are missing values

Audience exposure was driven by effective content distribution rather than posting volume alone.

Number of Postings



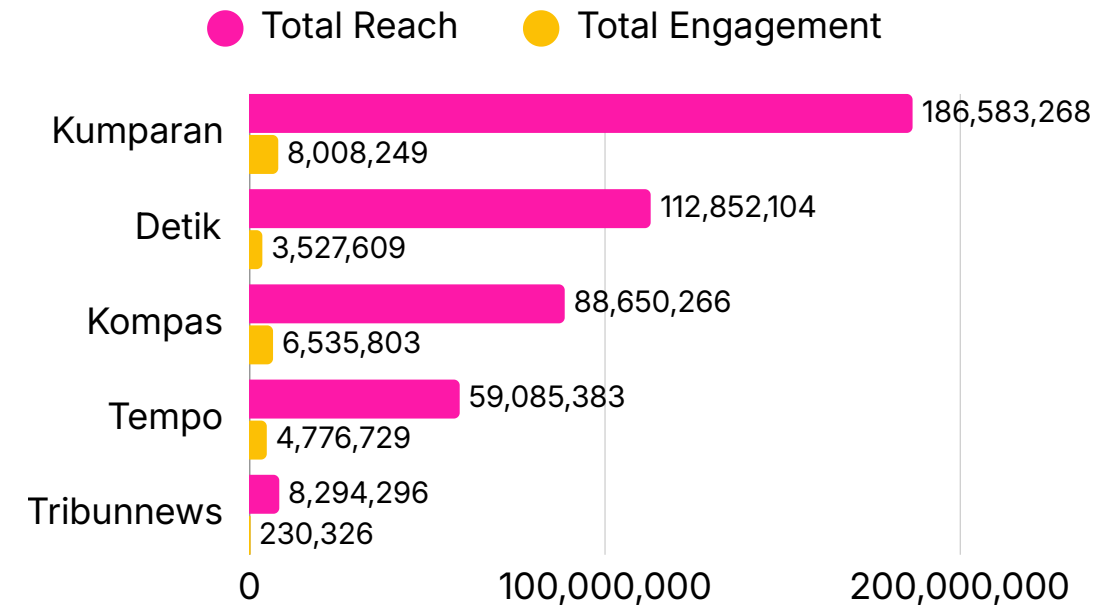
Dailyreach



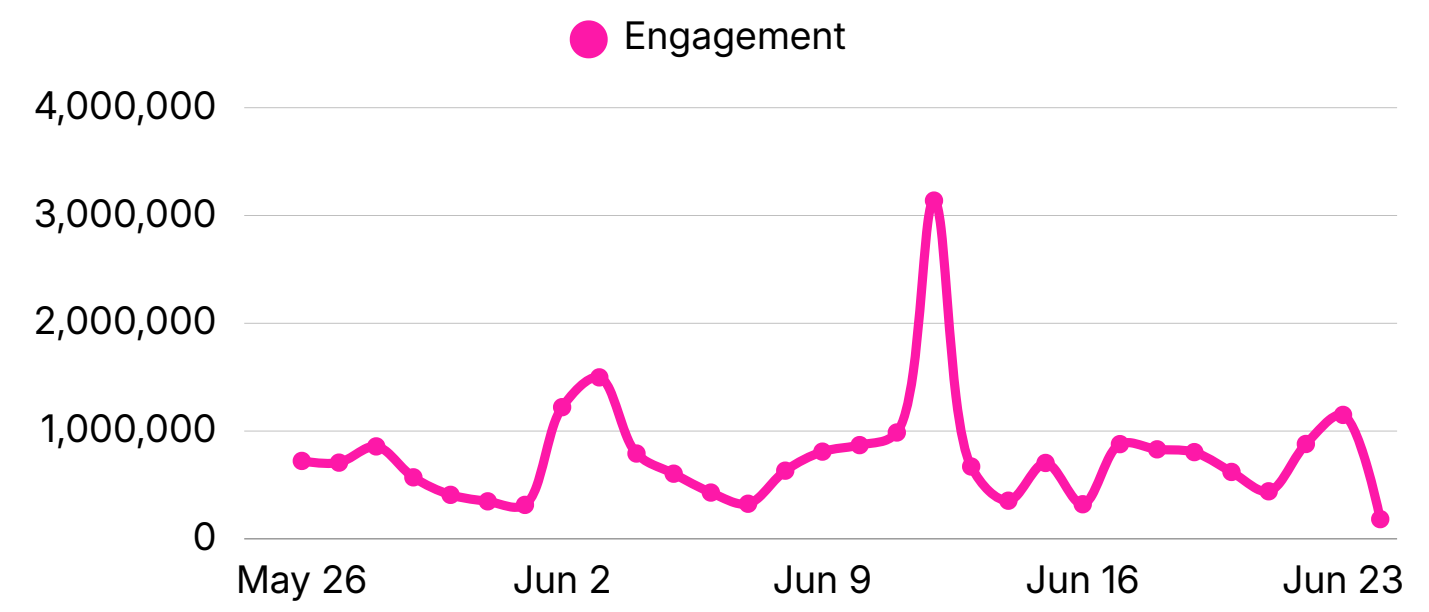
Key Insights

- Kumparan achieved the highest reach and engagement while producing the most content.
- Reach and engagement moved together over time, indicating strong alignment between them.

Totalreach & Engagement



Daily Engagement



Key Takeaway

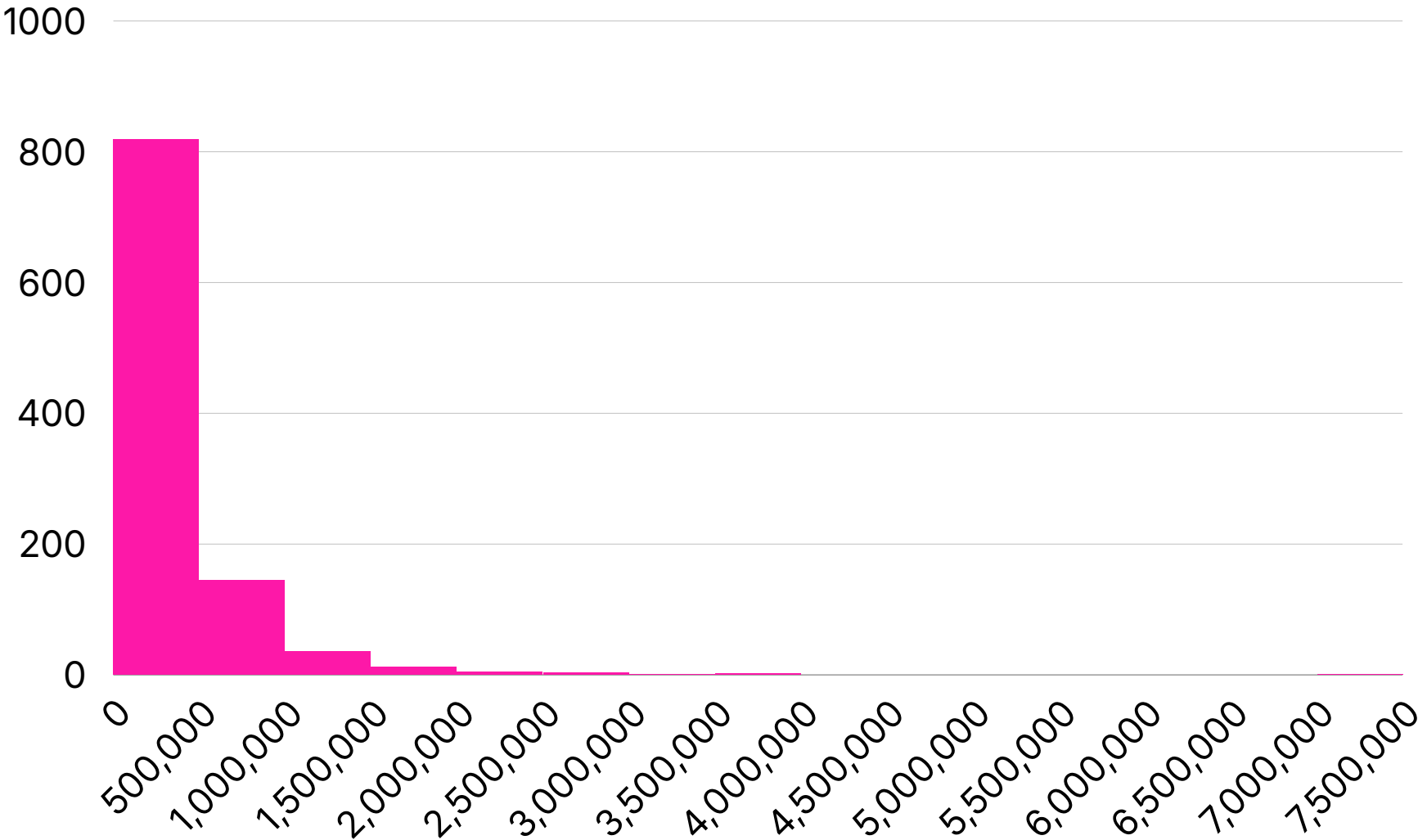
Content distribution effectiveness played a greater role in driving performance than content volume.

Key Impact

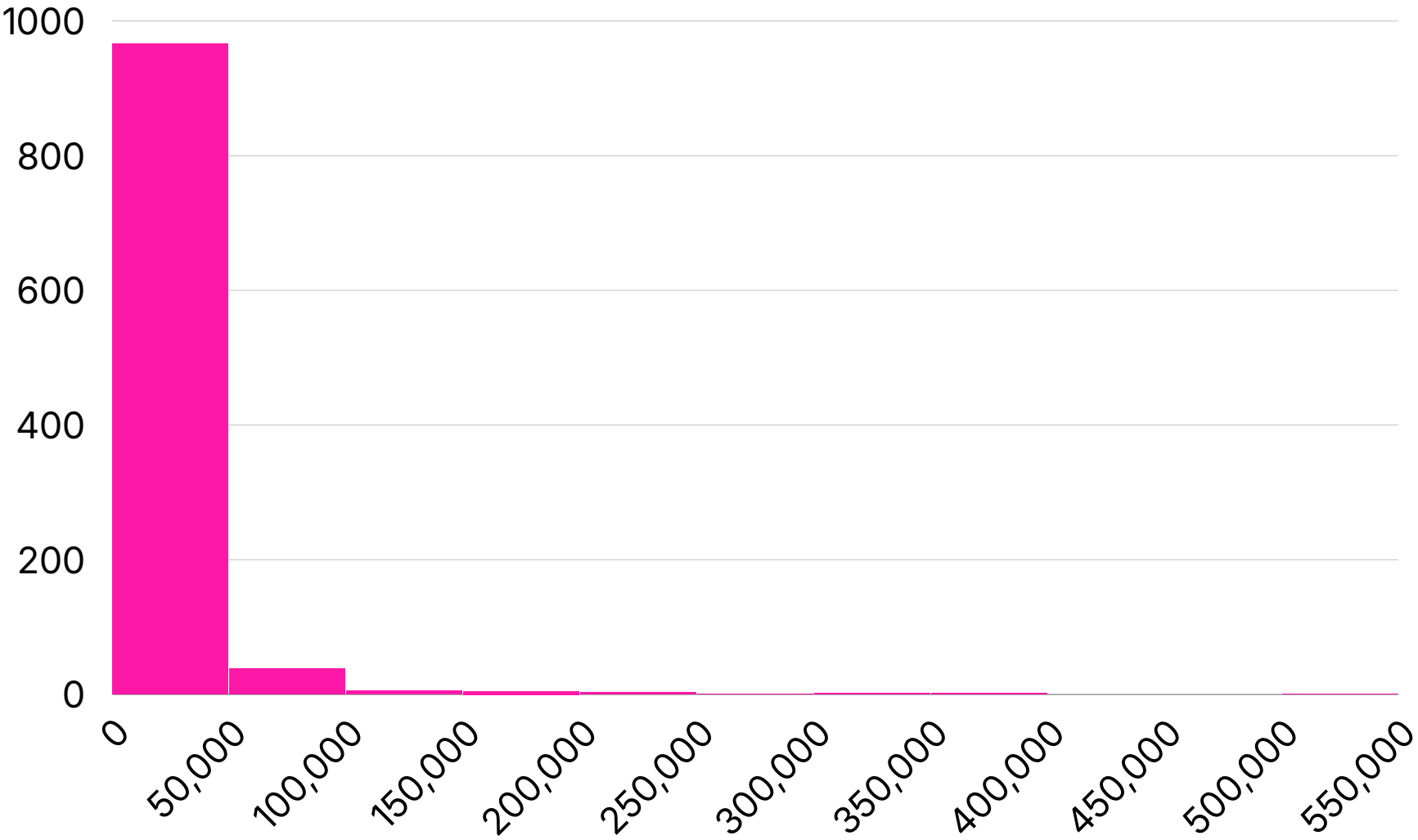
Improve content distribution strategies to maximize audience exposure and engagement.

Right-skewed performance distribution requires median-based evaluation to reduce outlier impact.

Reach Distribution



Engagement Distribution

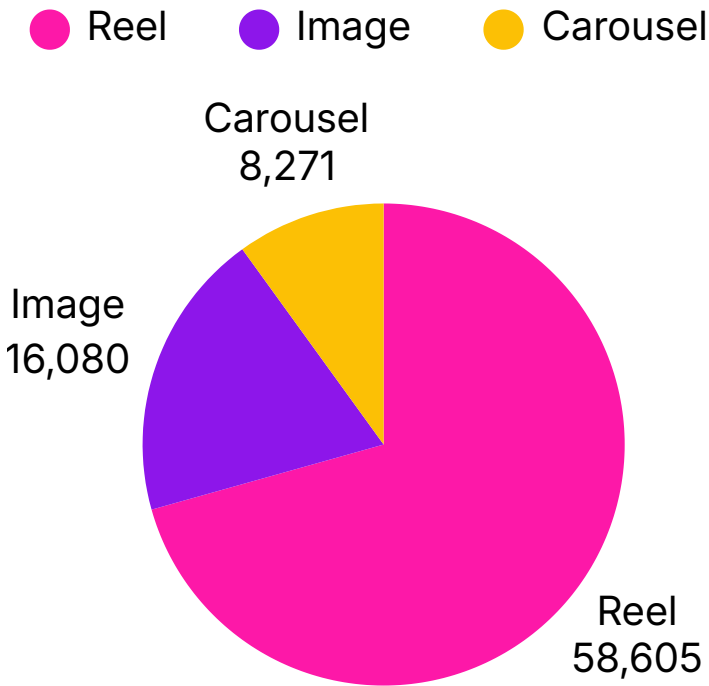


Key Insights

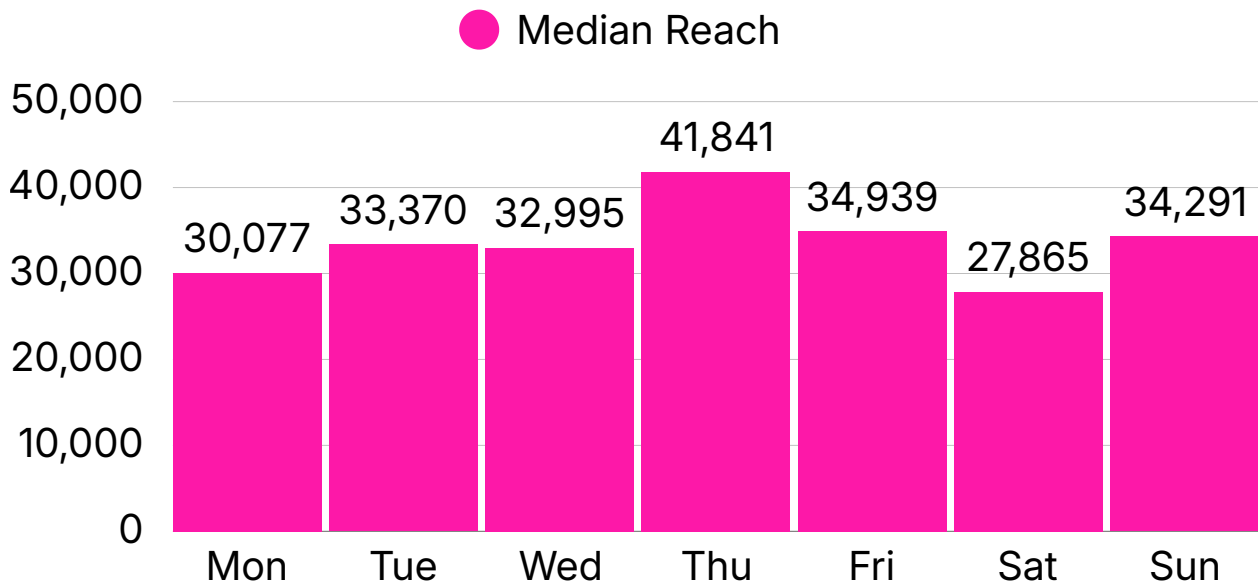
- ✓ The distributions of both reach and engagement are heavily right-skewed, indicating that performance is concentrated in a small number of high-performing posts rather than being evenly distributed.
- ✓ Given this skewness, the median provides a more reliable measure, as it minimizes distortion from viral outliers and better represents the typical content performance.

Reach was driven by video format, discoverability factors, and optimal posting timing.

Content Type by Medianreach



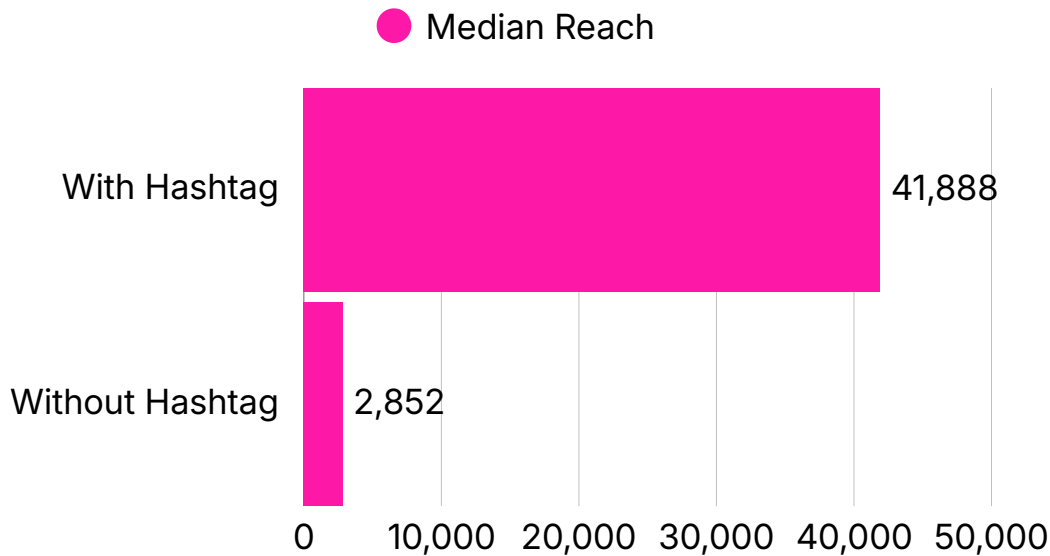
Prime Day



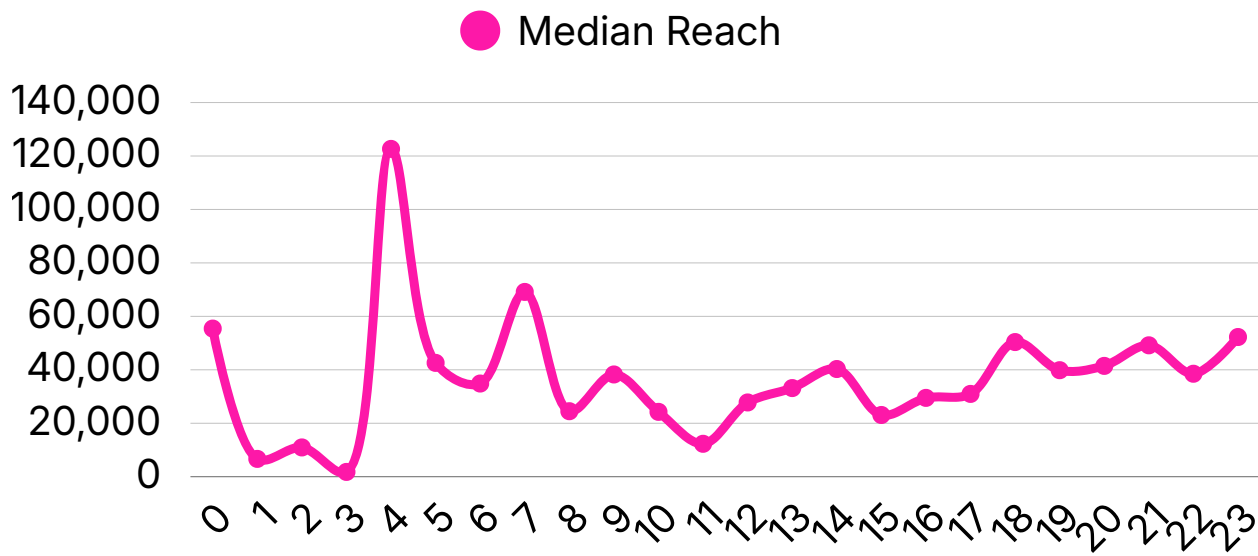
Key Insights

- Reels generated the highest reach, showing video content was more effective in expanding audience exposure.
- Posts with hashtags achieved higher reach, indicating improved content discoverability.
- Contents posted at Thursday and 04:00 showing the strongest performance while 01:00–03:00 underperformed.

Hashtag Usage



Prime Time



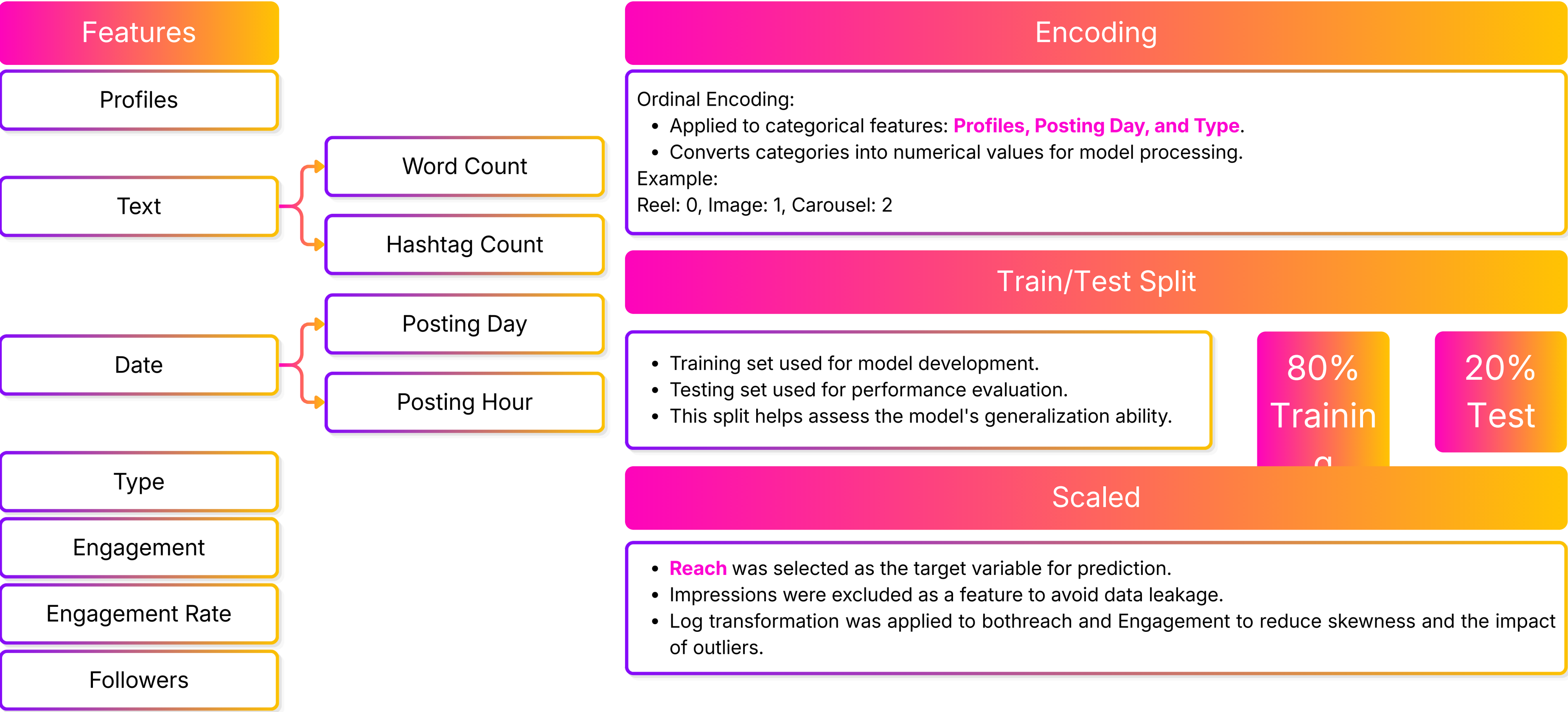
Key Takeaway

Content format, discoverability, and timing influenced audience reach more than posting alone.

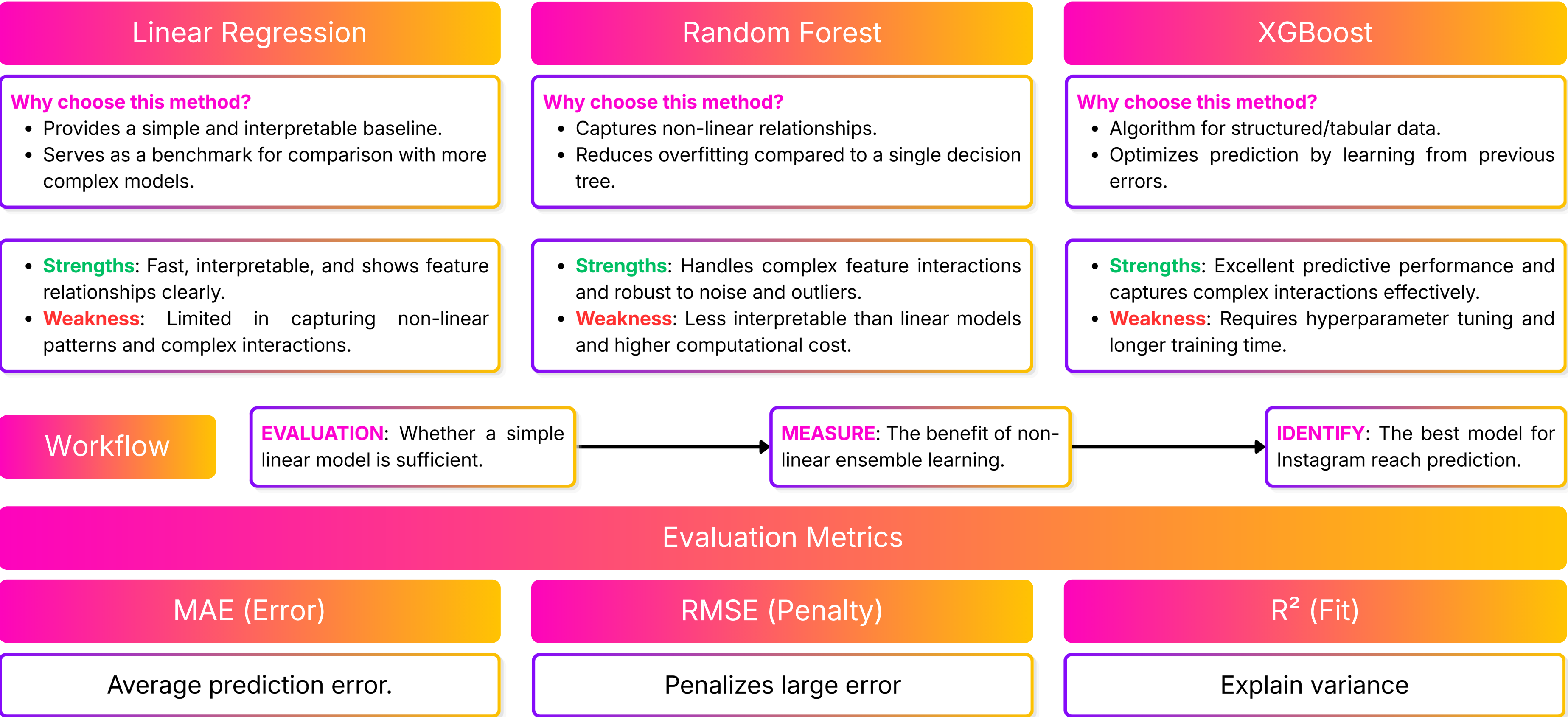
Key Impact

Optimize content strategy by prioritizing high-reach formats, relevant hashtags, and effective posting windows.

Data preprocessing prepared features and targets while reducing bias from skewed data and leakage.



Comparing Linear Regression, Random Forest, and XGBoost for each prediction.



Comparing Linear Regression, Random Forest, and XGBoost for each prediction.

Model	MAE	RMSE	R ²
Linear Regression	0.445307	0.579415	0.921846
Random Forest	0.305484	0.445369	0.953824
XGBoost	0.294515	0.430863	0.956783

Key Insights

 XGBoost outperformed Linear Regression and Random Forest with the highest R² (0.957) and lowest MAE (0.295) and RMSE (0.431).

 Tree-based models (Random Forest and XGBoost) performed better than Linear Regression, indicating nonlinear patterns influenced reach prediction.

Key Takeaway

Key Impact

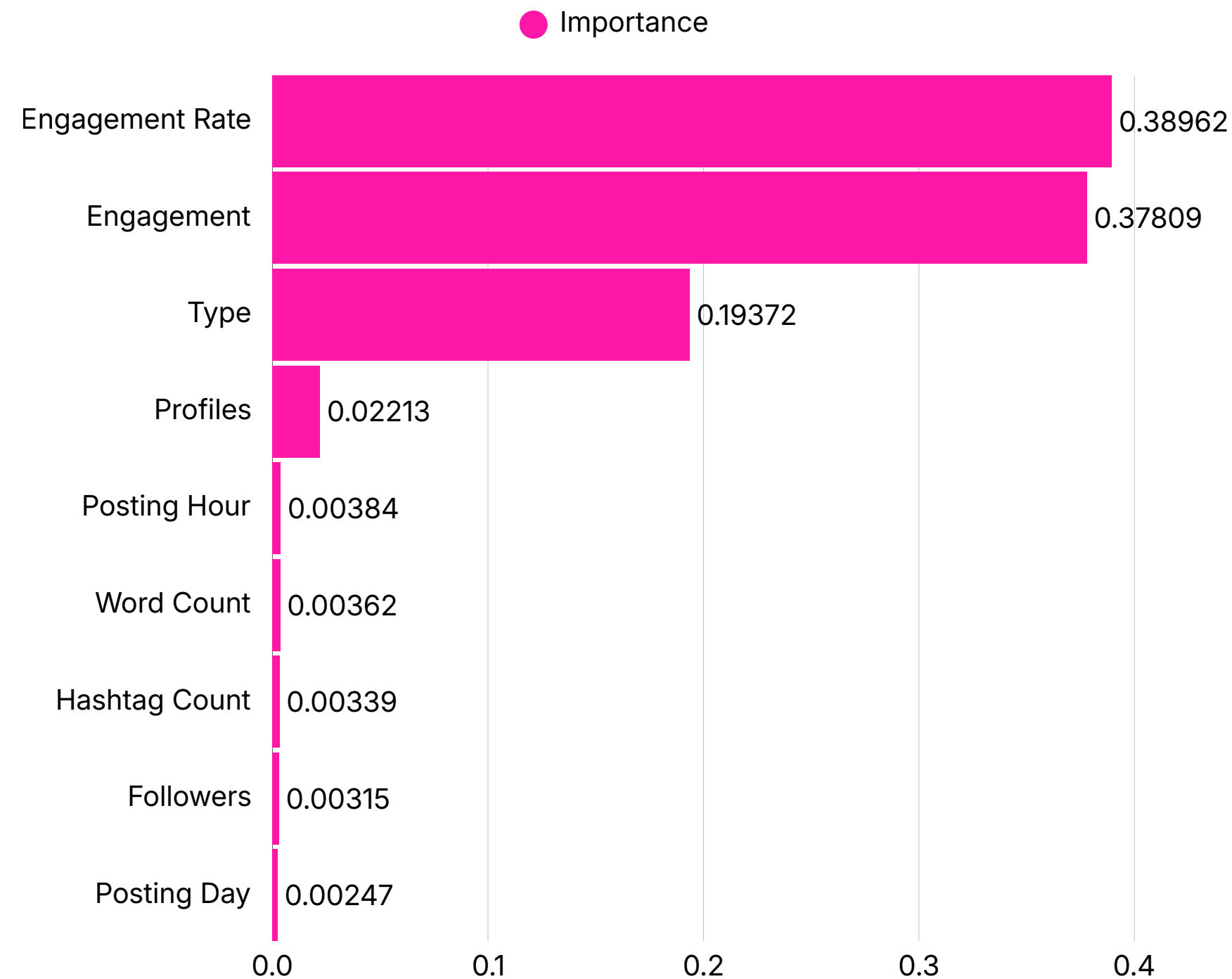
XGBoost was the most reliable model for predicting Instagram reach based on content characteristics.

Accurate reach prediction can support data-driven content planning and performance optimization.

Feature importance

Reach prediction was primarily driven by audience interactions and content format.

Feature Importance



Key Insights

- Engagement-related metrics were the strongest predictors of reach, indicating audience interactions provided the most valuable information.
- Content format influenced reach, but its impact was smaller compared to engagement factors.
- Account identity and posting-related features contributed minimally to prediction performance.

Key Takeaway

Reach was influenced more by content performance and audience response than publishing attributes.

Key Impact

Focus on creating engaging content and selecting effective formats to improve reach potential.

Business recommendations, actions, and expected impacts.

Recommendations	Key Finding	Recommended Action	Expected Impact
 Prioritize Reel Content	Reel content generated the highest median reach and dominated the top performing posts.	Increase the proportion of Reel posts, as Reels achieved the highest median reach among all content types.	Higher reach potential.
 Optimize Audience Engagement	Engagement-related metrics were the strongest predictors of reach .	Create content that encourages likes and comments, as Engagement and its rate were the strongest predictors of reach .	Improve organic reach through stronger audience interaction.
 Prioritize Posting on Prime Day and Time	Contents posted at Thursday and 04:00 showing the strongest performance.	Prioritize posting at weekdays and optimal hours, such as 04:00.	Higher reach potential.
 Evaluate Content Strategy Beyond Account Size	Reach were driven by engagement-metrics and content format, rather than account attributes.	Focus on improving content quality and engagement rather than relying solely on account attributes.	Higher reach potential.

LET'S CONNECT!



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